

Md Raihanul Haque Rahi

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SUMMARY

AI engineer working across agentic backends, real-time voice systems, and edge and on-device ML. Builds production Django/FastAPI systems, speech pipelines, and optimized computer-vision models, with peer-reviewed research in AI nutrition and open-source contributions to voice synthesis tooling.

EDUCATION

Rajshahi University of Engineering & Technology (RUET)
B.Sc. in Mechatronics Engineering | CGPA: 3.19/4.00

Rajshahi, Bangladesh
2022 – 2026

TECHNICAL SKILLS

Languages: Python, C/C++, Dart, Rust, Swift, SQL
Frameworks & Libraries: FastAPI, Django, Flutter, PyTorch, TensorFlow, MLX, LangChain, HuggingFace
Databases & Infra: PostgreSQL, Supabase, Redis, Firebase, Docker, WebRTC, LiveKit
AI / ML: Agentic LLM workflows, RAG, ASR/TTS/VAD, YOLO, stereo depth, TFLite, ONNX, MLX, quantization

EXPERIENCE

Intelsense AI
Backend Engineer

Apr 2024 – Present
Remote

- Own the backend architecture for an AI-powered SaaS platform, designing **Django/PostgreSQL** systems with a focus on modularity, reliability, and scale
- Architect production **RAG** services with structured ingestion and retrieval flows, resilient failure handling, and reliable product-facing APIs
- Build **OCR and ASR data-collection pipelines** and integrate LLM, speech, and document-processing capabilities into the web backend

Cortex Technologies
ML Engineer

Jan 2024 – Present
Contract Collaboration

- Own the ML engineering track for client projects sourced through a small freelance collaboration, building custom object detection pipelines, stereo depth estimation, and real-time edge-vision inference
- Built an automated object-detection dataset pipeline that uses **SAM 3** to generate bounding boxes from user-provided labels and exports annotations in **COCO format**
- Optimize and deploy models on resource-constrained hardware using **TFLite**, **ONNX**, and quantization techniques

Outlier / Remotasks
AI Trainer & LLM Data Quality Reviewer

Jul 2023 – Dec 2024
Remote

- Reviewed AI-generated math/code solutions and audited multilingual programming datasets for model reasoning and code-generation fine-tuning quality

PROJECTS

Gesprek | *FastAPI, PostgreSQL, Redis, Supabase, Docker, ElevenLabs, LiveKit*

Archived

- Built and deployed a **production-ready voice-agent platform** with configurable multi-agent workflows, Supabase auth, Redis-backed real-time TTS, subscriptions, and containerized services
- Engineered multi-tenant agent orchestration and independently scalable services; later archived after feasibility validation due to market and infrastructure constraints

StereoLite: Lightweight Stereo Depth Estimation | *PyTorch, ONNX · B.Sc. Thesis*

Active

- Designed a stereo depth model with **fewer than 3M parameters**, over **10× smaller** than DEFOM-Stereo (30M+), while using **4× less VRAM** (1.5 GB versus 6 GB) and reaching EPE 0.96 after 85,000 training steps
- Decoupled iterative multi-scale GRU refinement from a frozen DINOv2 backbone to reduce compute and memory costs while preserving depth-refinement quality across scales

Hishab: Agentic Expense Tracker | *Flutter, FastAPI, Supabase, SQLAlchemy*

May 2024 – Aug 2025

- Built a production Flutter+FastAPI app that converts Banglish natural-language prompts into structured financial records using **agentic tool-calling workflows**
- Implemented Supabase row-level-security auth, SMTP notification flows, SQLAlchemy persistence, and a companion web client

Diet Engine: AI Nutrition Coach | *Flutter, FastAPI, PyTorch, Firebase*

Feb – May 2024

- Built a Flutter client and FastAPI/PyTorch inference service for real-time food recognition, calorie tracking, personalized nutrition analysis, and Firebase authentication

PUBLICATION

- **“Diet Engine: A real-time food nutrition assistant system for personalized dietary guidance.”** *Food Chemistry Advances*, Elsevier, 2025. doi:10.1016/j.focha.2025.100978

ACHIEVEMENTS & ACTIVITIES

- **JAXA Kibo Robot Programming Challenge 2025: 1st place, Bangladesh regional; 4th place globally;** programmed Astrobees robots for zero-gravity navigation and AR object recognition aboard the ISS
- **Microsoft / VibeVoice:** Contributed package-import support to a 40k+ star emotional voice-synthesis repository
- **Leadership:** Codeforces rating **1280** (Pupil); led RUET coding club and taught C/C++ problem-solving to first-year students; runs Rahi 404 technical YouTube channel